

First-Order Logic in Artificial intelligence

Inference rules:

Commutative	$p \wedge q \iff q \wedge p$	$p \vee q \iff q \vee p$
Associative	$(p \wedge q) \wedge r \iff p \wedge (q \wedge r)$	$(p \vee q) \vee r \iff p \vee (q \vee r)$
Distributive	$p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$
Identity	$p \wedge T \iff p$	$p \vee F \iff p$
Negation	$p \vee \sim p \iff T$	$p \wedge \sim p \iff F$
Double Negative	$\sim(\sim p) \iff p$	
Idempotent	$p \wedge p \iff p$	$p \vee p \iff p$
Universal Bound	$p \vee T \iff T$	$p \wedge F \iff F$
De Morgan's	$\sim(p \wedge q) \iff (\sim p) \vee (\sim q)$	$\sim(p \vee q) \iff (\sim p) \wedge (\sim q)$
Absorption	$p \vee (p \wedge q) \iff p$	$p \wedge (p \vee q) \iff p$

Modus Ponens	$p \implies q$ p $\therefore q$	Modus Tollens	$p \implies q$ $\sim q$ $\therefore \sim p$
Elimination	$p \vee q$ $\sim q$ $\therefore p$	Transitivity	$p \implies q$ $q \implies r$ $\therefore p \implies r$
Generalization	$p \implies p \vee q$ $q \implies p \vee q$	Specialization	$p \wedge q \implies p$ $p \wedge q \implies q$
Conjunction	p q $\therefore p \wedge q$	Contradiction Rule	$\sim p \implies F$ $\therefore p$

First-Order Logic in Artificial intelligence

- PL is not sufficient to represent the complex sentences or natural language statements. The propositional logic has very limited expressive power. Consider the following sentence, which we cannot represent using PL logic.
- **"Some humans are intelligent", or**
- **"Sachin likes cricket."**
- To represent the above statements, PL logic is not sufficient, so we required some more powerful logic, such as first-order logic.

FOL

- First-order logic is another way of knowledge representation in artificial intelligence.
- First-order logic is also known as **Predicate logic or First-order predicate logic**. First-order logic is a powerful language that develops information about the objects in a more easy way and can also express the relationship between those objects.
- First-order logic (like natural language) does not only assume that the world contains facts like propositional logic but also assumes the following things in the world:
 - **Objects:** A, B, people, numbers, colors, wars, theories, squares, pits, wumpus,
 - **Relations:** It can be **unary relation such as:** red, round, is adjacent, **or n-any relation such as:** the sister of, brother of, has color, comes between
 - **Function:** Father of, best friend, third inning of, end of,
- first-order logic also has two main parts:**Syntax**
- **Semantics**

Syntax of First-Order logic:

- Basic Elements of First-order logic:

Constant	1, 2, A, John, Mumbai, cat,....
Variables	x, y, z, a, b,....
Predicates	Brother, Father, >,....
Function	sqrt, LeftLegOf,
Connectives	$\wedge$, $\vee$, $\neg$, $\Rightarrow$, $\Leftrightarrow$
Equality	$=$
Quantifier	$\forall$, $\exists$

Atomic sentences:

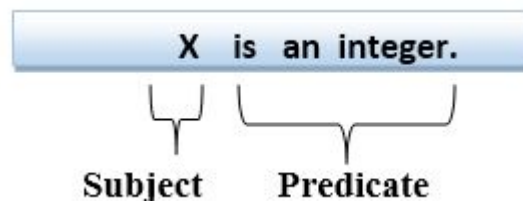
- Atomic sentences are the most basic sentences of first-order logic. These sentences are formed from a predicate symbol followed by a parenthesis with a sequence of terms.
- We can represent atomic sentences as **Predicate (term1, term2,, term n)**.
- **Example: Ravi and Ajay are brothers: $\Rightarrow$ Brothers(Ravi, Ajay).**

Complex Sentences:

Complex sentences are made by combining atomic sentences using connectives.

First-order logic statements can be divided into two parts:

- **Subject:** Subject is the main part of the statement.
- **Predicate:** A predicate can be defined as a relation, which binds two atoms together in a statement.



1. Sam was a man.
`man(sam)`

2. Cat is a animal.
`animal(cat)`

Quantifiers in First-order logic:

- A quantifier is a language element which generates quantification, and quantification specifies the quantity of specimen in the universe of discourse.
- These are the symbols that permit to determine or identify the range and scope of the variable in the logical expression. There are two types of quantifier:
 - **Universal Quantifier, (for all, everyone, everything)**
 - **Existential quantifier, (for some, at least one).**



$\forall x P(x)$

Universal means all elements satisfy 'is wearing a hat'



$\exists x P(x)$

Existential means at least one does

Universal quantifier ($\forall$)

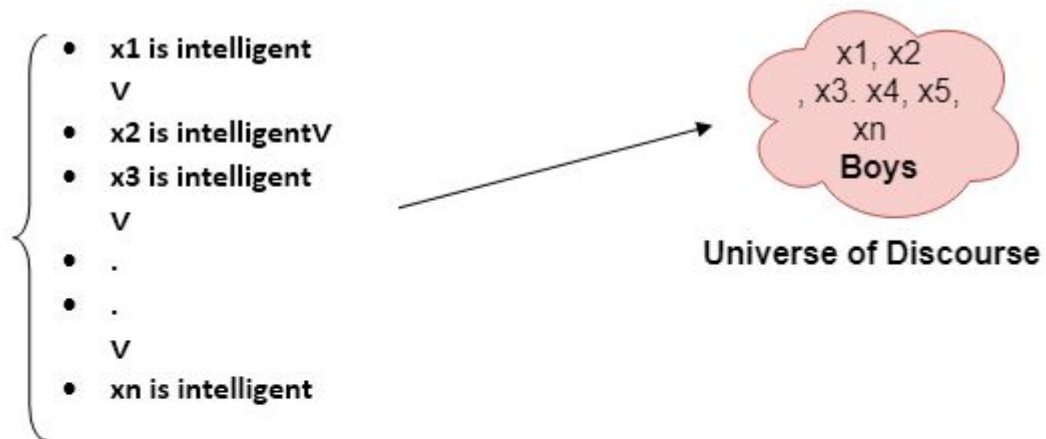
- $\forall x$: means “for all” x $(\forall)$
- It is used to represent phrase “for all”.
- It says that something is true for all possible values of a variable.
 - In universal quantifier we use implication “ $\rightarrow$ ”.
- Ex. “John hates everyone”

$\forall x$: hates(John , x)

Existential quantifiers

- Existential quantifiers are the type of quantifiers, which express that the statement within its scope is true for at least one instance of something.
- It is denoted by the logical operator $\exists$, which resembles as inverted E.
- In Existential quantifier we always use AND or Conjunction symbol ($\wedge$).
- “some people like reading and hence they gain good knowledge”
- $\exists x: \{[\text{person}(x) \wedge \text{like}(x, \text{reading})] \rightarrow \text{gain}(x, \text{knowledge})\}$

Some boys are intelligent.



So in short-hand notation, we can write it as:

$\exists x: \text{boys}(x) \wedge \text{intelligent}(x)$

EXAMPLES

1. All birds fly.

$$\forall x (Bird(x) \rightarrow Fly(x))$$

(For every x, if x is a bird then x flies.)

2. Every man respects his parent.

Interpretation: every man respects each of his parents.

$$\forall x (Man(x) \rightarrow \forall y (Parent(y, x) \rightarrow Respect(x, y)))$$

(For every x, if x is a man then for every y, if y is a parent of x then x respects y.)

3. Some boys play cricket.

$$\exists x (Boy(x) \wedge Play(x, cricket))$$

(There exists at least one x such that x is a boy and x plays cricket.)

4. Not all students like both Mathematics and Science.

Two equivalent FOPL forms:

- Negation of universal form:

$$\neg \forall x (Student(x) \rightarrow (Like(x, Math) \wedge Like(x, Science)))$$

- Existential equivalent (often clearer):

$$\exists x (Student(x) \wedge (\neg Like(x, Math) \vee \neg Like(x, Science)))$$

(There exists a student who fails to like at least one of Math or Science.)

- Connection between $\forall$ and $\exists$

- “everyone dislikes garlic”

- $\forall x: \neg \text{like} (x, \text{garlic})$

□ This can be also said as:

- “there does not exists someone who likes garlic”

- $\neg \exists x: \text{like} (x, \text{garlic})$

Quantifier Scope

- Switching the order of universal quantifiers *does not* change the meaning:
 - $(\forall x)(\forall y)P(x,y) \leftrightarrow (\forall y)(\forall x) P(x,y)$
- Similarly, you can switch the order of existential quantifiers:
 - $(\exists x)(\exists y)P(x,y) \leftrightarrow (\exists y)(\exists x) P(x,y)$

- Shows the use of predicate logic as a way of representing knowledge.
- 1. Marcus was a man.
- 2. Marcus was a Pompeian.
- 3. All Pompeians were Romans.
- 4. Caesar was a ruler.
- 5. All Pompeians were either loyal to Caesar or hated him.
- 6. Everyone is loyal to someone.
- 7. People only try to assassinate rulers they are not loyal to.
- 8. Marcus tried to assassinate Caesar.

The facts described by well-formed formulas (wffs)

as follows:

1. Marcus was a man.

- $\text{man}(\text{Marcus})$

2. Marcus was a Pompeian.

- $\text{Pompeian}(\text{Marcus})$

3. All Pompeians were Romans.

- $\forall x: \text{Pompeian}(x) \rightarrow \text{Roman}(x)$

4. Caesar was a ruler.

- $\text{ruler}(\text{Caesar})$

5. All Pompeians were either loyal to Caesar or hated him.

- inclusive-or

- $\forall x: \text{Roman}(x) \rightarrow \text{loyalto}(x, \text{Caesar}) \vee \text{hate}(x, \text{Caesar})$

- exclusive-or

- $\forall x: \text{Roman}(x) \rightarrow (\text{loyalto}(x, \text{Caesar}) \wedge \neg \text{hate}(x, \text{Caesar})) \vee (\neg \text{loyalto}(x, \text{Caesar}) \wedge \text{hate}(x, \text{Caesar}))$

6. Everyone is loyal to someone.

- $\forall x: \exists y: \text{loyalto}(x, y)$

7. People only try to assassinate rulers they are not loyal to.

- $\forall x: \forall y: \text{person}(x) \wedge \text{ruler}(y) \wedge \text{tryassassinate}(x, y) \rightarrow \neg \text{loyalto}(x, y)$

8. Marcus tried to assassinate Caesar.

- $\text{tryassassinate}(\text{Marcus}, \text{Caesar})$

Connections between All and Exists

We can relate sentences involving $\forall$ and $\exists$ using De Morgan's laws:

$$(\forall x) \neg P(x) \leftrightarrow \neg (\exists x) P(x)$$

$$\neg (\forall x) P \leftrightarrow (\exists x) \neg P(x)$$

$$(\forall x) P(x) \leftrightarrow \neg (\exists x) \neg P(x)$$

$$(\exists x) P(x) \leftrightarrow \neg (\forall x) \neg P(x)$$

Quantified inference rules

- Universal instantiation
 - $\forall x P(x) \therefore P(A)$
- Universal generalization
 - $P(A) \wedge P(B) \dots \therefore \forall x P(x)$
- Existential instantiation
 - $\exists x P(x) \therefore P(F)$ $\leftarrow$ skolem constant F
- Existential generalization
 - $P(A) \therefore \exists x P(x)$

Universal instantiation (a.k.a. universal elimination)

□ Universally quantified sentence

- All students in the AI class are smart
 $\forall x \text{ Student}(\text{AICourse}, x) \Rightarrow \text{Smart}(x)$

□ Intuitively, x can be anything:

- $\text{Student}(\text{AICourse}, \text{Tom}) \Rightarrow \text{Smart}(\text{Tom})$
- $\text{Student}(\text{AICourse}, \text{Chair}) \Rightarrow \text{Smart}(\text{Chair})$
- $\text{Student}(\text{AICourse}, \text{LeftLeg}(\text{John})) \Rightarrow \text{Smart}(\text{LeftLeg}(\text{John}))$

□ Formally:

$$\frac{\forall x \quad S}{\text{Subst}(\{x/p\}, S)}$$

- x is replaced with p (ground term) in S , and the quantifier is removed.

Existential instantiation (a.k.a. existential elimination)

□ Existentially Quantified sentence

- There is smart student in the AI class $\exists x \text{ Student}(\text{AICourse}, x) \wedge \text{Smart}(x)$

□ Intuitively, x must name something. But what?

- Can we conclude: $\text{Student}(\text{AICourse}, \text{Tom}) \wedge \text{Smart}(\text{Tom})$???
- No! The sentence might not be true for Tom !

□ Instead, Use a Skolem constant and draw the conclusion, Formally

$$\frac{\exists x \quad S}{\text{Subst}(\{x/K\}, S)}$$

- K is called a Skolem constant and is a completely new symbol you created.

Existential generalization (a.k.a. existential introduction)

- If $P(c)$ is true, then $(\exists x) P(x)$ is inferred.

Example: Let's say that,

"Priyanka got good marks in English."

"Therefore, someone got good marks in English."

- All instances of the given constant symbol are replaced by the new variable symbol
- Note that the variable symbol cannot already exist anywhere in the expression

Universal Generalization:

- Universal generalization is a valid inference rule which states that if premise $P(c)$ is true for any arbitrary element c in the universe of discourse, then we can have a conclusion as $\forall x P(x)$.
- Example: Let's represent, $P(c)$:
- "A byte contains 8 bits", so for $\forall x P(x)$
- "All bytes contain 8 bits.",
- it will also be true.

Translating English to FOL

Every gardener likes the sun.

You can fool some of the people all of the time.

You can fool all of the people some of the time.

All purple mushrooms are poisonous.

No purple mushroom is poisonous.

Clinton is not tall.

Translating English to FOL

Every gardener likes the sun.

$$\forall x \text{ gardener}(x) \rightarrow \text{likes}(x, \text{Sun})$$

You can fool some of the people all of the time.

$$\exists x \forall t \text{ person}(x) \wedge \text{time}(t) \rightarrow \text{can-fool}(x, t)$$

You can fool all of the people some of the time.

$$\forall x (\text{person}(x) \rightarrow \exists t (\text{time}(t) \wedge \text{can-fool}(x, t)))$$

Equivalent

All purple mushrooms are poisonous.

$$\forall x (\text{mushroom}(x) \wedge \text{purple}(x)) \rightarrow \text{poisonous}(x)$$

No purple mushroom is poisonous.

$$\forall x (\text{mushroom}(x) \wedge \text{purple}(x)) \rightarrow \neg \text{poisonous}(x)$$

Equivalent

Clinton is not tall.

$$\neg \text{tall}(\text{Clinton})$$

Unification

- It's a matching procedure that compares two literals and discovers whether there exists a set of substitutions that can make them identical.

UNIFY(knows(Richard, x), knows(Richard, John))

Here, $\Psi_1 = \text{knows(Richard, x)}$, and $\Psi_2 = \text{knows(Richard, John)}$

$S_0 \Rightarrow \{ \text{knows(Richard, x); knows(Richard, John)} \}$

SUBST $\theta = \{ \text{John/x} \}$

$S_1 \Rightarrow \{ \text{knows(Richard, John); knows(Richard, John)} \}$, **Successfully Unified.**

Unifier: {John/x}.

Resolution

- Resolution is used, if there are various statements are given, and we need to prove a conclusion of those statements. Unification is a key concept in proofs by resolutions. Resolution is a single inference rule which can efficiently operate on the **conjunctive normal form or clausal form**.
- **Clause**: Disjunction of literals (an atomic sentence) is called a **clause**. It is also known as a unit clause.
- **Conjunctive Normal Form**: A sentence represented as a conjunction of clauses is said to be **conjunctive normal form or CNF**.

Resolution algorithm

Conversion of facts into first-order logic.

Convert FOL statements into CNF

Negate the statement which needs to prove (proof by contradiction)

Draw resolution graph (unification).

John likes all kind of food.

Apple and vegetable are food

Anything anyone eats and not killed is food.

Anil eats peanuts and still alive

Harry eats everything that Anil eats.

**Prove by resolution that:
John likes peanuts.**

- a. $\forall x: \text{food}(x) \rightarrow \text{likes}(\text{John}, x)$
 - b. $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
 - c. $\forall x \forall y: \text{eats}(x, y) \wedge \neg \text{killed}(x) \rightarrow \text{food}(y)$
 - d. $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil}).$
 - e. $\forall x : \text{eats}(\text{Anil}, x) \rightarrow \text{eats}(\text{Harry}, x)$
 - f. $\forall x: \neg \text{killed}(x) \rightarrow \text{alive}(x)$
 - g. $\forall x: \text{alive}(x) \rightarrow \neg \text{killed}(x)$
 - h. $\text{likes}(\text{John}, \text{Peanuts})$
- } **added predicates.**

○ **Eliminate all implication ($\rightarrow$) and rewrite**

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- $\forall x \forall y \neg [\text{eats}(x, y) \wedge \neg \text{killed}(x)] \vee \text{food}(y)$
- $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
- $\forall x \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
- $\forall x \neg [\neg \text{killed}(x)] \vee \text{alive}(x)$
- $\forall x \neg \text{alive}(x) \vee \neg \text{killed}(x)$
- $\text{likes}(\text{John}, \text{Peanuts})$

○ **Move negation ($\neg$) inwards and rewrite**

- $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- $\forall x \forall y \neg \text{eats}(x, y) \vee \text{killed}(x) \vee \text{food}(y)$
- $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
- $\forall x \neg \text{eats}(\text{Anil}, x) \vee \text{eats}(\text{Harry}, x)$
- $\forall x \neg \text{killed}(x) \vee \text{alive}(x)$
- $\forall x \neg \text{alive}(x) \vee \neg \text{killed}(x)$
- $\text{likes}(\text{John}, \text{Peanuts})$

Rename variables or standardize variables

- a. $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- b. $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- c. $\forall y \forall z \neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
- d. $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
- e. $\forall w \neg \text{eats}(\text{Anil}, w) \vee \text{eats}(\text{Harry}, w)$
- f. $\forall g \neg \text{killed}(g) \vee \text{alive}(g)$
- g. $\forall k \neg \text{alive}(k) \vee \neg \text{killed}(k)$
- h. $\text{likes}(\text{John}, \text{Peanuts})$.

Eliminate existential instantiation quantifier by elimination.

In this step, we will eliminate existential quantifier $\exists$, and this process is known as **Skolemization**. But in this example problem since there is no existential quantifier so all the statements will remain same in this step.

o **Drop Universal quantifiers.**

In this step we will drop all universal quantifier since all the statements are not implicitly quantified so we don't need it.

- a. $\neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- b. $\text{food}(\text{Apple})$
- c. $\text{food}(\text{vegetables})$
- d. $\neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
- e. $\text{eats}(\text{Anil}, \text{Peanuts})$
- f. $\text{alive}(\text{Anil})$
- g. $\neg \text{eats}(\text{Anil}, w) \vee \text{eats}(\text{Harry}, w)$
- h. $\text{killed}(g) \vee \text{alive}(g)$
- i. $\neg \text{alive}(k) \vee \neg \text{killed}(k)$
- j. $\text{likes}(\text{John}, \text{Peanuts})$.

- a. $\forall x \neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
- b. $\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetables})$
- c. $\forall y \forall z \neg \text{eats}(y, z) \vee \text{killed}(y) \vee \text{food}(z)$
- d. $\text{eats}(\text{Anil}, \text{Peanuts}) \wedge \text{alive}(\text{Anil})$
- e. $\forall w \neg \text{eats}(\text{Anil}, w) \vee \text{eats}(\text{Harry}, w)$
- f. $\forall g \neg \text{killed}(g) \vee \text{alive}(g)$
- g. $\forall k \neg \text{alive}(k) \vee \neg \text{killed}(k)$
- h. $\text{likes}(\text{John}, \text{Peanuts})$.

